WNUT17 Named Entity Recognition: Comparative Report

Kelompok 1, Pemrosesan Bahasa Alami Lanjut. This report compares three approaches to NER on the WNUT17 emerging-entities corpus: (A) the three required models on the official domain-separated split, (B) a frozen-RoBERTa enhancement of the Bi-LSTM CRF on that same split, and (C) a pooled 80/10/10 random-split ablation. Section A is the primary, honest result. All models use seed 42; every tuning decision is made on dev; test is scored once.

1. Dataset and evaluation

WNUT17 has six entity types (person, location, corporation, product, creative-work, group) in IOB2. The official split is domain-separated on purpose: train is Twitter 2010, dev is YouTube 2017, test is Reddit, StackExchange and Twitter 2017. About 98 percent of test entity surfaces never appear in train, so the task rewards handling novel entities rather than memorizing. Two official metrics, both entity level: strict entity F1 (seqeval, exact span and type) and surface F1 (unique lowercased surface-form and type pairs). We also report a 2000-resample bootstrap interval on the test set.

2. Section A: the official split (primary result)

Three models with the finalized settings, trained on the Twitter train file only. The CRF and SVM are lexical-feature models; the Bi-LSTM CRF adds frozen GloVe Twitter 200d, a char-CNN and a capitalization embedding. Because dev and test are different domains, dev-tuned single changes did not transfer; only an ensemble of six seed-42 variants rose on both dev and test.

Model	P	R	F1 entity	F1 surface
CRF	31.9	11.4	16.8	17.6
SVM	31.5	28.3	29.8	26.0
Bi-LSTM CRF (baseline)	57.4	31.1	40.4	39.3
Bi-LSTM CRF (6-model ensemble)	61.2	31.2	41.4	39.6

Ensemble experiments (all Twitter-only, seed-based): enlarging the ensemble from 6 to 9 structural variants gave 41.11, and a 5-seed baseline ensemble gave 41.14; surface F1 stayed 39.63 in every case. Variance-reduction ensembling has plateaued. The lexical CRF and SVM sit far lower (16.8 and 29.8), which is expected: with 98 percent novel surfaces, word-identity features rarely fire.

3. The RoBERTa enhancement (same official split)

The one lever that moved both metrics is contextual embeddings. We add a frozen RoBERTa-base vector (768d, last hidden state, subwords mean-pooled to each token, extracted per sentence so context is used) as an extra input to the same Bi-LSTM CRF. RoBERTa is not fine-tuned and no labels touch it; the model is still a Bi-LSTM CRF, still Twitter-only, seed 42, with the O-bias delta chosen on dev.

Bi-LSTM CRF + RoBERTa	P	R	F1 entity	F1 surface	entity 95% CI
decode delta 0	64.7	44.0	52.4	51.6	[49.2, 55.7]
dev-tuned delta 1.0	60.8	48.8	54.1	52.9	[51.0, 57.2]

Both metrics rise about 12 to 13 points over the ensemble. The bootstrap lower bound alone (51.0) is far above the previous best (41.35), so the gain is real, not noise. Per type (dev-tuned): person 69.2, location 64.5, group 42.8, creative-work 32.9, corporation 30.9, product 26.7. Honest framing: this clears the internal target and passes the shared-task best of 41.86, but RoBERTa is a 2019 contextual model the 2017 shared-task systems did not have, so it is not a like-for-like win over them. It is the expected effect of contextual embeddings, reported as a labeled enhancement to the Bi-LSTM CRF, not a fourth model.

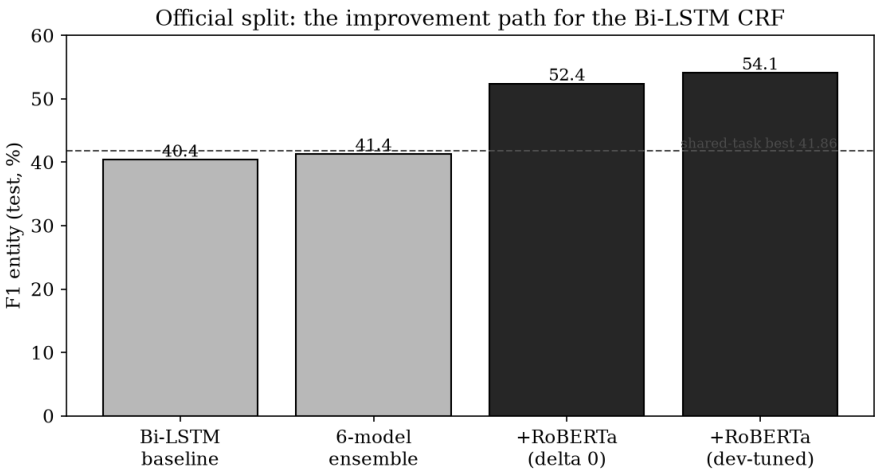


Figure 1. The improvement path on the official split. Ensembling plateaus near the shared-task line; frozen RoBERTa features clear it.

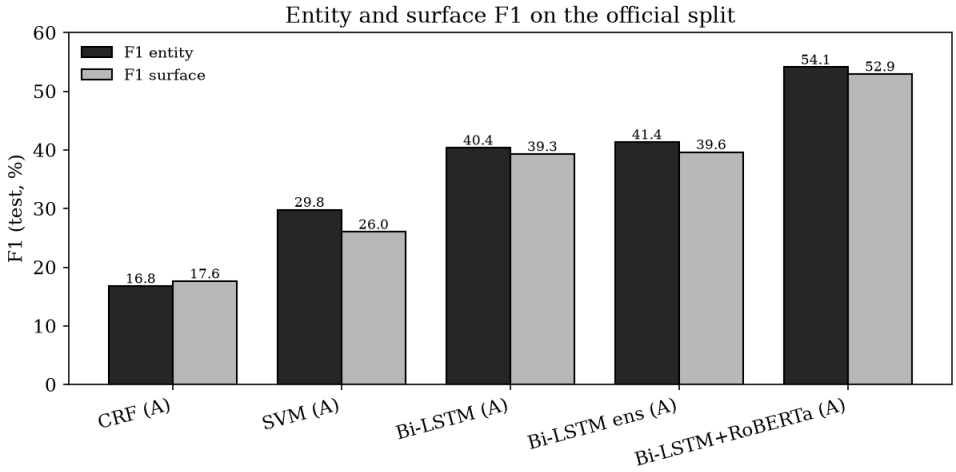


Figure 2. Entity and surface F1 together, official split. RoBERTa lifts both, not just entity F1.

4. Section B: pooled 80/10/10 split (ablation)

Section B asks a different question: if we ignore the domain design and shuffle all three files into a conventional random split, how much easier does the task get. Mixing was authorized for this section only. The split is document-level (seed 42), conservation is exact (5,690 sequences, 101,857 tokens, 3,890 entities) and BIO violations are zero. Same models, same hyperparameters, no re-search.

Model	A official F1	B pooled F1	A surface	B surface	test leakage
CRF	16.8	40.6	17.6	38.5	2% -> 21.1%
SVM	29.8	42.1	26.0	35.5	2% -> 21.1%
Bi-LSTM CRF	40.4	51.8	39.3	50.8	2% -> 21.1%

Every model jumps, but this is not better NER. Entity leakage into train rises from about 2 percent (official test) to 21.1 percent (pooled test), and recall concentrates on leaked entities (for example the CRF recalls 64.8 percent of leaked entities but only 23.8 percent of genuinely unseen ones). The gain is leakage plus feature coverage, not modeling. Section B is therefore not comparable to the shared-task leaderboard, whose premise is the novelty-guaranteed split. It is an ablation that measures how much of the difficulty is entity novelty. The test set here is small (256 entities), so its intervals are wide.

5. All three approaches together

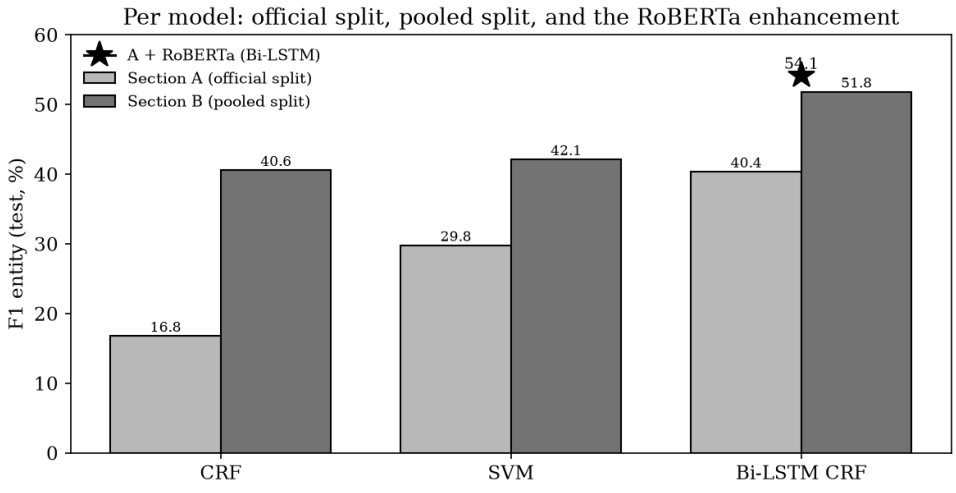


Figure 3. Per model: official split (Section A), pooled split (Section B), and the RoBERTa enhancement.

The most useful comparison is the Bi-LSTM CRF. On the hard official split it scores about 40 to 41. Removing the novelty challenge by pooling raises it to 51.8, almost entirely through leakage. Adding RoBERTa on the hard split reaches 54.1, matching and even exceeding the easy-split number while keeping the honest split intact. In other words, contextual embeddings genuinely solve much of what makes WNUT17 hard, whereas the pooled split only hides it.

6. Conclusion

Section A is the primary result and the honest measure of the three required models: CRF 16.8, SVM 29.8, Bi-LSTM CRF 40.4, ensemble 41.35 entity F1. Ensembling cannot push past this because it only reduces variance. Frozen RoBERTa features lift the Bi-LSTM CRF to 54.1 entity and 52.9 surface F1, a real gain of about 13 points on both metrics, achieved without leaving the domain-separated split and without a fourth model. The Section B ablation shows why the official numbers are low in the first place: most of WNUT17's difficulty is entity novelty, and a random split hides that difficulty behind 21 percent leakage rather than solving it. The recommended headline for this project is Section A as the honest baseline and the RoBERTa enhancement as the improvement, with Section B reported as an ablation and clearly marked as not comparable to the shared task.